



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DR.MUNIRATHINAM FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

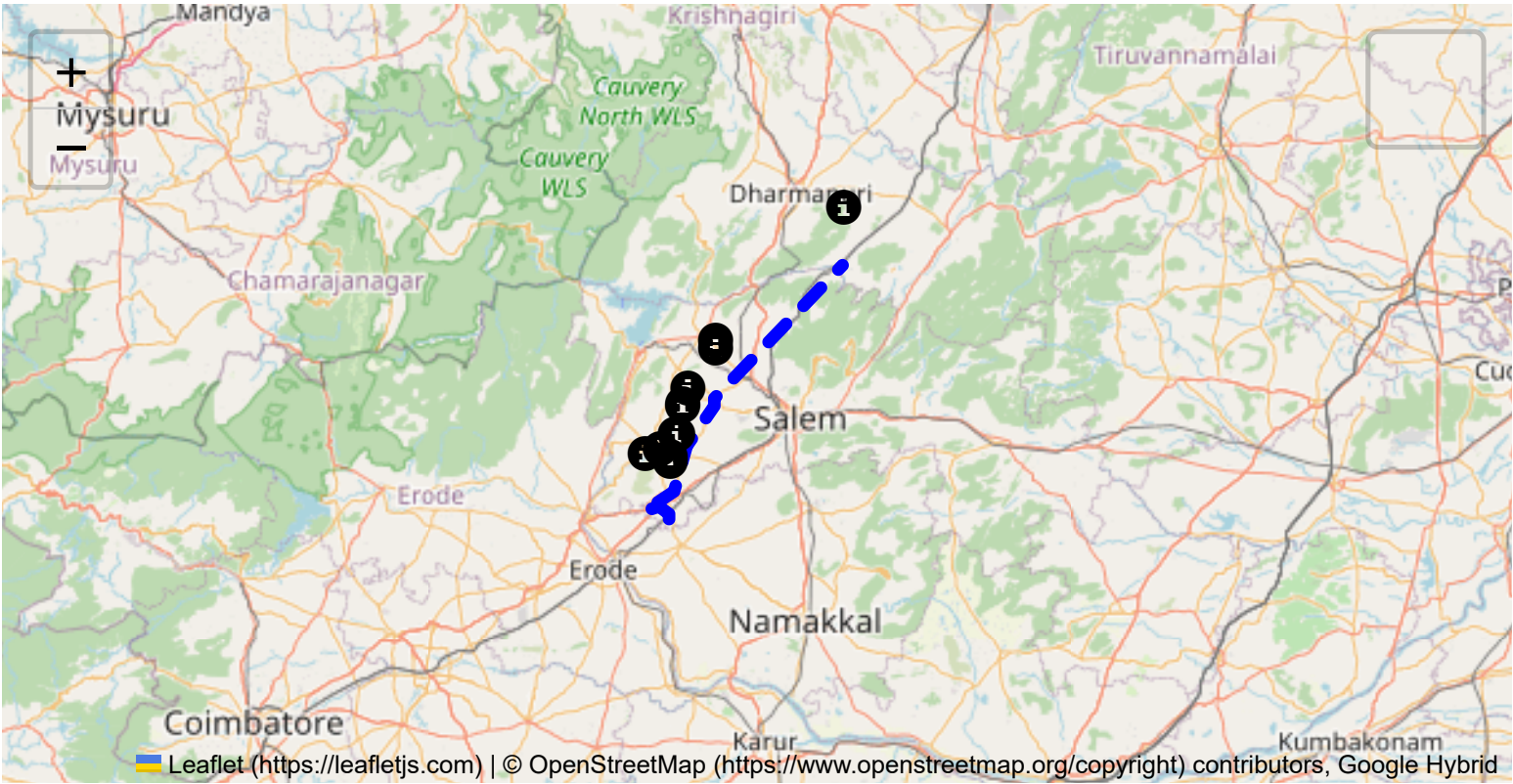
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

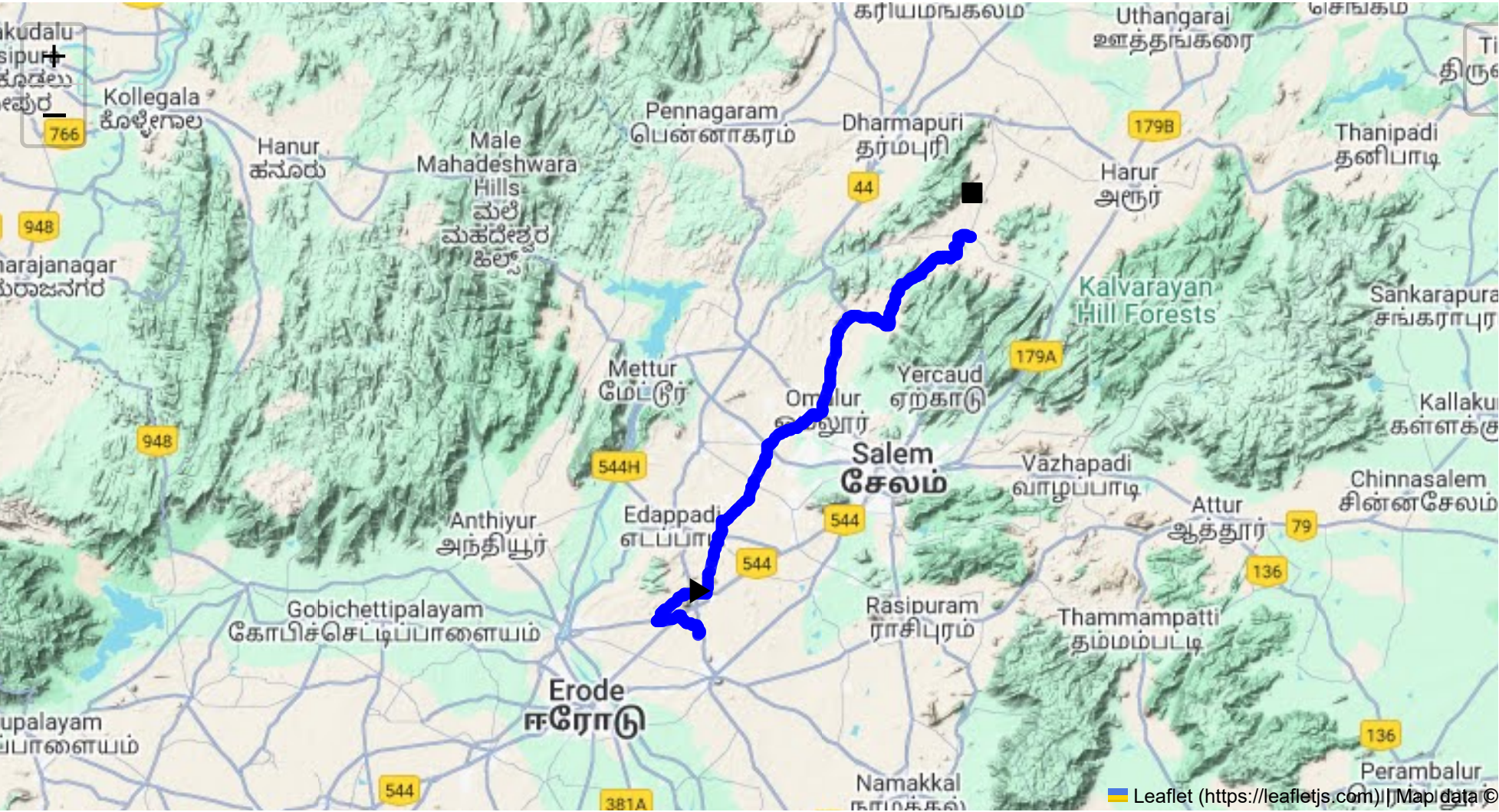
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 97.94 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (11.979316, 78.252426)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route spans approximately 97.94 kilometers, starting at CVQF+23W, Sangagiri, Tamil Nadu, and concluding at 154, Salem Main Rd, Bommidi, Tamil Nadu. This journey involves traversing several state highways, mainly utilizing SH 86, and passes through various towns including Sankari, Padaiveedu, Erumaipatti, and Tharamangalam.

2. Typical Weather Conditions and Potential Hazards:

The region experiences a tropical climate with hot summers, considerable rainfall during the monsoon season (June to September), and mild winters. Potential weather-related hazards include:

- **Heavy Rainfall:** This can lead to slippery roads and reduced visibility during the monsoon.
- **Visibility Issues:** Fog can be prevalent during early mornings, especially in winter months.

3. Traffic Patterns and Congestion-Prone Areas:

Traffic congestion tends to occur during peak hours, typically from 8:00 to 10:00 AM and 6:00 to 8:00 PM. Congestion-prone areas along this route include:

- **Sankari and Sankari West:** Due to their proximity to industrial areas.
- **Tharamangalam and Erumaipatti:** Residential areas with regular local traffic.

4. Road Quality and Infrastructure:

Road quality varies, with certain sections like SH 86 being better maintained. However, expect uneven surfaces and potholes in rural sections or near smaller towns. Infrastructure like signage and lighting may be inadequate, particularly in less populated stretches.

5. Suggestions for Alternative Routes for Emergencies:

In emergencies, consider using the NH 544 which provides a quicker bypass for many southern sections. Additionally, local roads connecting major towns can be used as alternative bypasses.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

Transporting hazardous materials requires adherence to specific regulations, such as:

- Valid permits for hazardous materials.
- Appropriate labeling and documentation.
- Restrictions on operating times, mostly during flood warnings or official travel advisories.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

There have been occasional incidents involving truck overturns or spills, particularly during monsoon season due to slippery roads. Regular enforcement of weight limits is advised to prevent road damage and accidents.

8. Environmental Considerations and Sensitive Areas:

The route traverses through agricultural zones, thus extra caution is advised to avoid contamination. There are no major protected environmental areas, but local farmlands must be respected.

9. Analysis of Communication Coverage, Noting Potential Dead Zones:

Mobile coverage is generally decent along the state highways. However, expect signal drops in rural and less populated areas, particularly between smaller towns on the route.

10. Estimated Emergency Response Times for Different Route Segments:

- **Urban Areas (e.g., Tharamangalam, Sankari):** Approximately 15-30 minutes due to proximity to emergency services.
- **Rural Sections:** 30-60 minutes; response times may be longer due to road conditions and distance from emergency stations.

11. Overall Summary of Risk Assessment:

The route presents moderate risks primarily due to weather conditions, traffic congestion, and variable road quality. Safety measures, such as adherence to traffic rules, speed limits, and careful navigation during adverse weather conditions, are advised. Continual monitoring of weather and traffic updates can further aid in mitigating risks. Effective communication strategies need to be in place to manage potential dead zones.

By ensuring compliance with local regulations and maintaining safety protocols, the route remains manageable for the transport of hazardous materials. However, continuous evaluation and adaptation to emerging conditions are necessary to ensure ongoing safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57035, 77.90223	30 KM/Hr
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.58279, 77.90658	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	High	11.69595, 77.97114	15 KM/Hr
21	Turn	Medium	11.71386, 77.99552	30 KM/Hr
22	Turn	Medium	11.71735, 78.00019	30 KM/Hr
23	Turn	Medium	11.72765, 78.02291	30 KM/Hr
24	Turn	Medium	11.73786, 78.03505	30 KM/Hr
25	Turn	High	11.73652, 78.03866	15 KM/Hr
26	Turn	Medium	11.74257, 78.04708	30 KM/Hr
27	Turn	Medium	11.77998, 78.05786	30 KM/Hr
28	Turn	Medium	11.86915, 78.08382	30 KM/Hr
29	Turn	High	11.87099, 78.08358	15 KM/Hr
30	Turn	High	11.87110, 78.08383	15 KM/Hr
31	Turn	Medium	11.87072, 78.08439	30 KM/Hr
32	Turn	Medium	11.87088, 78.08556	30 KM/Hr
33	Turn	High	11.87246, 78.08704	15 KM/Hr
34	Turn	Medium	11.86898, 78.12074	30 KM/Hr
35	Turn	Medium	11.86495, 78.12530	30 KM/Hr
36	Turn	High	11.85820, 78.13580	15 KM/Hr
37	Turn	Medium	11.86044, 78.13937	30 KM/Hr
38	Turn	Medium	11.86652, 78.13879	30 KM/Hr
39	Turn	Medium	11.88411, 78.14478	30 KM/Hr
40	Turn	Medium	11.88540, 78.14474	30 KM/Hr
41	Turn	Medium	11.89899, 78.14930	30 KM/Hr
42	Turn	Medium	11.90581, 78.15292	30 KM/Hr
43	Turn	High	11.90760, 78.15205	15 KM/Hr
44	Turn	Medium	11.91226, 78.15594	30 KM/Hr
45	Turn	Medium	11.91312, 78.15616	30 KM/Hr
46	Turn	Medium	11.91519, 78.15837	30 KM/Hr
47	Turn	High	11.92454, 78.17073	15 KM/Hr
48	Turn	Medium	11.92448, 78.17340	30 KM/Hr
49	Turn	Medium	11.92820, 78.18040	30 KM/Hr
50	Turn	Medium	11.92797, 78.18165	30 KM/Hr
51	Turn	Medium	11.93585, 78.18730	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
52	Turn	Medium	11.95116, 78.20647	30 KM/Hr
53	Turn	Medium	11.95200, 78.20703	30 KM/Hr
54	Turn	High	11.95258, 78.20848	15 KM/Hr
55	Turn	Medium	11.95173, 78.21017	30 KM/Hr
56	Turn	High	11.95221, 78.21173	15 KM/Hr
57	Turn	Medium	11.95165, 78.21276	30 KM/Hr
58	Turn	Medium	11.95228, 78.21413	30 KM/Hr
59	Turn	High	11.95220, 78.21522	15 KM/Hr
60	Turn	Medium	11.95288, 78.21603	30 KM/Hr
61	Turn	Medium	11.95264, 78.22467	30 KM/Hr
62	Turn	High	11.95178, 78.22600	15 KM/Hr
63	Turn	Medium	11.95847, 78.23410	30 KM/Hr
64	Turn	Medium	11.96509, 78.23292	30 KM/Hr
65	Turn	High	11.96652, 78.23302	15 KM/Hr
66	Blind Spot	Blind Spot	11.96679, 78.23211	10 KM/Hr
67	Turn	High	11.97329, 78.23713	15 KM/Hr
68	Turn	High	11.97765, 78.23556	15 KM/Hr
69	Turn	High	11.98240, 78.24290	15 KM/Hr
70	Turn	High	11.98205, 78.24333	15 KM/Hr
71	Turn	High	11.98346, 78.24637	15 KM/Hr
72	Turn	Medium	11.97927, 78.25177	30 KM/Hr
73	Turn	Medium	11.97934, 78.25210	30 KM/Hr
74	Turn	High	11.97951, 78.25228	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
32	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
33	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
34	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
35	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
36	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
37	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium
38	hospital	Raja Hospital, Dharmapuri	11.9792908, 78.2509482	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57035, 77.90223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57041, 77.90221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58279, 77.90658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69595, 77.97114



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73786, 78.03505



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87099, 78.08358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87110, 78.08383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87072, 78.08439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87088, 78.08556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87246, 78.08704



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86898, 78.12074



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95116, 78.20647



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97329, 78.23713



Coordinates: 11.97765, 78.23556



Coordinates: 11.98240, 78.24290



Coordinates: 11.98205, 78.24333



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98346, 78.24637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97927, 78.25177



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.97934, 78.25210



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97951, 78.25228